Student placement prediction and skill analysis

Problem Statement

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Students often lack clarity about their placement chances and the skills required to get placed. Most rely on general advice instead of data-driven insights, making it difficult to identify strengths and weaknesses.

There is a need for a system that can predict placement outcomes and provide personalized suggestions to help students improve their chances.

project explantion

This system predicts whether a student will get placed based on their academic performance, skills, and experience.
The system takes student data such as CGPA, aptitude score, communication skills, and internship experience as input. It then processes this data using trained machine learning models to generate predictions.

The output includes:

- Placement prediction (Placed / Not Placed)
- Placement probability (e.g., 78%)
- Personalized suggestions to improve placement chances

Algorithms

1. Logistic Regression

- A statistical classification model used to predict binary outcomes (Placed / Not Placed)
- Calculates the probability of placement using a sigmoid function
- Works well for understanding the relationship between input features and output

Role in Project:

Used to generate placement probability (e.g., 78%)

2. Decision Tree

- A tree-structured model that splits data based on feature conditions
- Makes decisions using rules like "if CGPA > 7 then..."
- Captures non-linear relationships between features

Role in Project:

Used for decision-making logic and understanding how features affect placement

3. Random Forest

- An ensemble model that combines multiple decision trees
- Each tree is trained on different data samples (bagging technique)
- Reduces overfitting and improves prediction accuracy
- Provides feature importance scores

Role in Project:

- Improves overall accuracy
- Identifies most important features for placement

Ensemble Approach

- Combines predictions from all models
- Uses probability-based averaging (soft voting concept)
- Produces more stable and reliable results than a single model

Model Performance

The performance of the models was evaluated using standard classification metrics:

- **Logistic Regression** → 85% Accuracy
- **Decision Tree** → 88.33% Accuracy
- **Random Forest** → 88.33% Accuracy
- **Ensemble Model** → 90% Accuracy

Evaluation metrics used

1. Accuracy

- Measures overall correctness of the model
- Ratio of correct predictions to total predictions

2. Precision

- Measures how many predicted "placed" students are actually placed
- Important to reduce false positives

3. Recall

- Measures how many actual placed students are correctly identified
- Important to avoid missing potential candidates

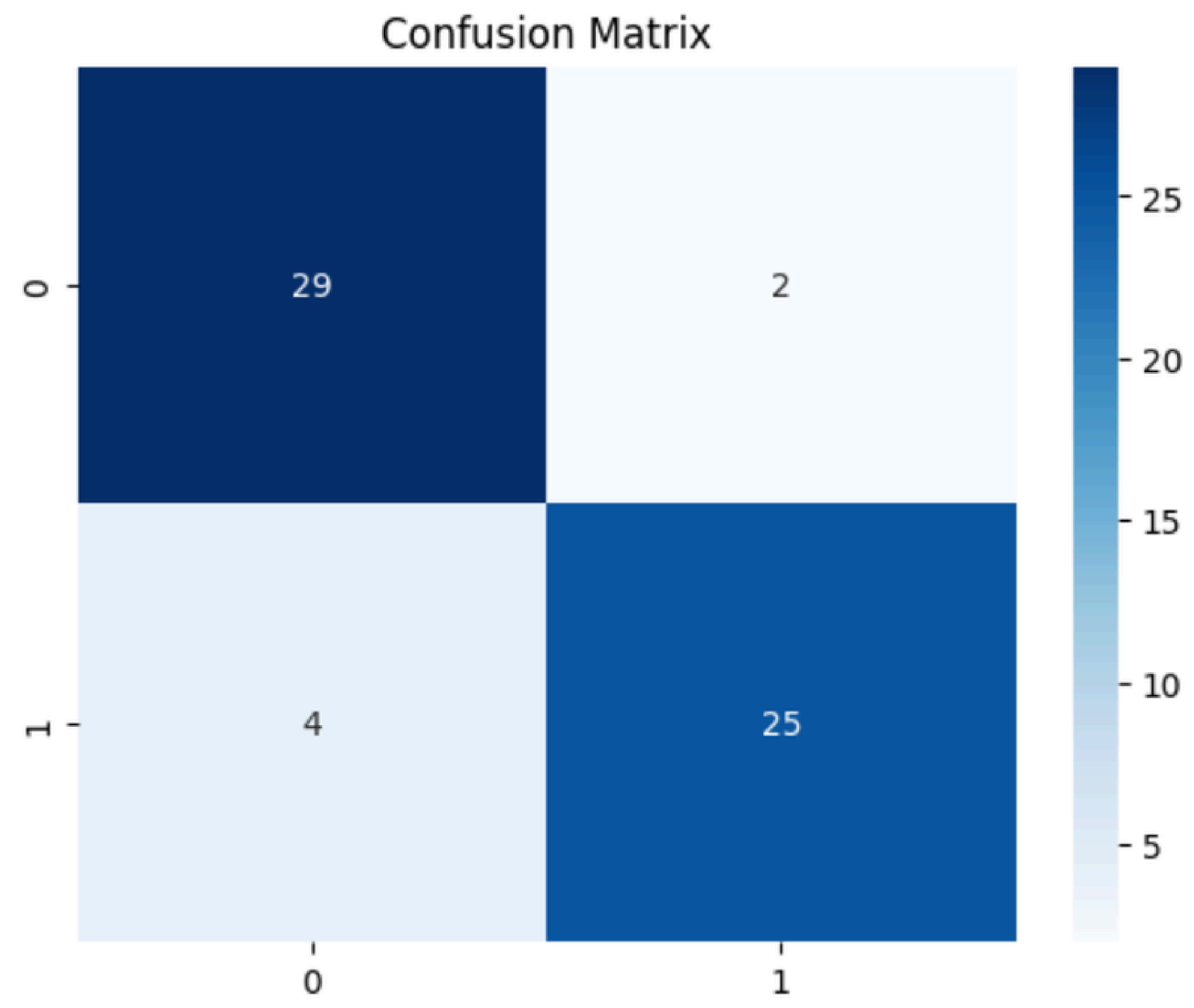
4. F1-Score

- Harmonic mean of precision and recall
- Gives balanced performance measure

5. Confusion Matrix

- Shows correct and incorrect predictions
- Includes: True Positive,True NEgative,False Positive,False Negatve

OUTPUTS

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 MODEL PERFORMANCE
Logistic Regression : 83.33%
Decision Tree       : 81.67%
Random Forest       : 88.33%
Ensemble Model ⭐    : 91.67%

 Classification Report (Ensemble)

              precision    recall  f1-score   support

           0       0.88      0.97      0.92        31
           1       0.96      0.86      0.91        29

    accuracy                           0.92        60
   macro avg       0.92      0.91      0.92        60
weighted avg       0.92      0.92      0.92        60
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Prediction

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 ENTER YOUR DETAILS:

Enter CGPA (0-10): 8.0
Enter Projects Completed (1-5): 1
Enter Internship (0=No, 1=Yes): 0
Enter Communication Skills (0-10): 7.5
Enter Aptitude Score (0-100): 75
Enter 12th Percentage (0-10): 8.0
Enter Previous Semester (0-10): 7.8
Enter Extra Activities (0-10): 8.5

 GENERATING IMPROVEMENT PLAN...


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 FINAL RESULT
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Prediction: PLACED ✅
Placement Probability: 75.59%

 Skills to Focus:
- Projects
- Internship
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 IMPROVEMENT PLAN
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Step 1: Improve Projects
   Current: 1.00
   Target : 3.59
   New Probability: 83.17%
   Gain: +7.57%

Step 2: Improve Internship
   Current: 0.00
   Target : 0.43
   New Probability: 77.77%
   Gain: +2.18%
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Conclusion

- Machine Learning provides more accurate and reliable predictions compared to traditional methods
- The system helps students understand their strengths and weaknesses
- Personalized suggestions guide students to improve their placement chances
- Using an ensemble approach improves overall model performance

Thank You